

B 2.4.2 Linear and Binary Search



B 2.4.2 Construct and trace algorithms to implement a linear search and a binary search for data retrieval.

Dive into the world of search algorithms. This section explores linear and binary search, comparing their efficiency and guiding you in choosing the right one for your needs. Learn how to implement and trace these algorithms in Java and Python, and discover how data organization impacts search efficiency in real-world scenarios.

Lesson objectives

- Students can describe the steps involved in linear and binary search algorithms.
- Students can implement linear and binary search algorithms in Java and Python.
- Students can trace the execution of linear and binary search algorithms for given inputs.
- Students will be able to compare the efficiency of linear and binary search algorithms using Big O notation.
- Students can select the appropriate search algorithm based on data search requirements and efficiency considerations.

IB ATL Skills

Thinking: Critical thinking, creative thinking

Research: Information literacy

Communication: Communication, collaboration

Self-Management: Organization, time management

Lesson Planning

Part 1: Linear Search (1 hour)

Engage (10 minutes)

Real-world scenario: Present a scenario like searching for a specific book on a bookshelf without organisation. Ask students how they would approach this task.

Discussion: Guide the discussion towards checking each item individually, which is the essence of linear search.

Explore (20 minutes)

Introduce linear search: Explain the algorithm as a simple, sequential search that checks each element in a list until the target value is found or the end of the list is reached.

Explain (15 minutes)

Implementation: Demonstrate how to implement linear search in both Java and Python. Provide code examples and explain the logic clearly.

Tracing: Work through a few examples with different inputs, tracing the algorithm's execution step by step.

Elaborate (10 minutes)

Efficiency Analysis: Discuss the time complexity of linear search ($O(n)$). Explain that the worst-case scenario is when the target is at the end of the list or not present.

Applications: Mention situations where linear search might be suitable (e.g., small datasets, unsorted data).

Evaluate (5 minutes)

Quick Quiz: Ask short-answer questions or provide code snippets for students to analyze and determine the time complexity of linear search.

Part 2: Binary Search (1 hour)

Engage (10 minutes)

Real-world scenario: Present a scenario like searching for a word in a dictionary. Ask students how they efficiently find the word without checking every single page.

Discussion: Guide the discussion towards the strategy of repeatedly halving the search space, which is the core idea behind binary search.

Explore (20 minutes)

Introduce binary search: Explain the algorithm as an efficient search method that works on sorted data by repeatedly dividing the search interval in half.

Explain (15 minutes)

Implementation: Demonstrate how to implement binary search in both Java and Python. Provide code examples and explain the logic, including calculating the middle index and adjusting the search interval.

Tracing: Work through examples with different inputs, tracing the execution of the algorithm and showing how the search interval narrows down.

Elaborate (10 minutes)

Efficiency Analysis: Discuss the time complexity of binary search ($O(\log n)$). Explain why it's much more efficient than linear search for large datasets.

Applications: Mention real-world applications of binary search (e.g., searching in databases, finding elements in sorted arrays).

Evaluate (5 minutes)

Worksheet: Provide a worksheet with code snippets or algorithm descriptions for students to analyze and determine the time complexity of the binary search.

Lesson Resources

Presentations

Java

[Link to presentation](#)

Python

[Link to presentation](#)

eBook

[B 2.4.2 Java](#)

[B 2.4.2 Python](#)

Worksheets

There are two worksheets for each language, which students can complete at the end of the presentation.

The programming exercises require them to modify the search method to ensure a deeper understanding, but it may be worth getting them to implement the simpler version during the lesson explanation so they internalise the process a little first.

[B 2.4.2 Worksheet Linear Search in Java](#)

[B 2.4.2 Worksheet Binary Search in Java](#)

[B 2.4.2 Worksheet Linear Search in Python](#)

[B 2.4.2 Worksheet Linear Search in Python updated](#)

[B 2.4.2 Worksheet Binary Search in Python](#)

The answer sheet provides the answers for all the worksheets. A table of contents at the top of the page facilitates navigation to the correct answer key.

B 2.4.2 Worksheet Answer Key

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